补充：

密度算符 $\hat{\rho}(t)$.

(a) 纯态密度算符

$$\rho_{纯}(t) = |\psi\rangle\langle\psi|$$

(b) 混态密度算符

$$\rho_{混}(t) = \sum_{i=1}^{s} P_i |\psi_i\rangle\langle\psi_i|$$

$$\sum_{i=1}^{s} P_i = 1 \quad , \quad s > 1$$

(c) 统一描述

$$\hat{\rho}(t) = \sum_{i=1}^{s} P_i |\psi_i\rangle\langle\psi_i|$$

$$\sum_{i=1}^{s} P_i = 1 \quad , \quad s \geq 1$$

$$Tr[\hat{\rho}(t)] = 1 \quad , \quad Tr[\hat{\rho}^2(t)] \leq 1$$

补充：

密度算符：$\hat{\rho}(t)$

- $\hat{\rho}(t) = \hat{U}(t\ t_0)\ \hat{\rho}(t_0)\ \hat{U}^{\dagger}(t\ t_0)$

- $\hat{\rho}^2(t) = \hat{U}(t\ t_0)\ \hat{\rho}(t_0)\ \hat{U}^{\dagger}(t\ t_0)\ \hat{U}(t\ t_0)\ \hat{\rho}(t_0)\ \hat{U}^{\dagger}(t\ t_0)$

$= \hat{U}(t\ t_0)\ \hat{\rho}^2(t_0)\ \hat{U}^{\dagger}(t\ t_0)$

- $Tr[\hat{\rho}^2(t)] = Tr[\hat{U}(t\ t_0)\ \hat{\rho}^2(t_0)\ \hat{U}^{\dagger}(t\ t_0)]$

$= Tr[\hat{\rho}^2(t_0)\ \hat{U}^{\dagger}(t\ t_0)\ \hat{U}(t\ t_0)]$

$= Tr[\hat{\rho}^2(t_0)]$. $\Rightarrow$

$\frac{d}{dt} Tr[\hat{\rho}^2(t)] = 0$

§2.7 复合系统的密度算符和密度矩阵

- 由两个子系统构成的复合系统

第一个子系统基矢：$\{|u_i(1)\rangle\}$，N维

〃 算符：$\hat{A}(1)$ → 正交、完备

第二个子系统基矢：$\{|v_k(2)\rangle\}$，M维

〃 算符：$\hat{B}(2)$

- 复合系统基矢：$\{|u_i(1)\rangle \otimes |v_k(2)\rangle\}$

N×M 维

〃 算符：$\hat{A}(1) \otimes \hat{B}(2)$

- 内积：$(\langle u_j| \otimes \langle v_l|)(|u_i\rangle \otimes |v_k\rangle)$

$= \langle u_j|u_i\rangle\langle v_l|v_k\rangle = \delta_{ji}\delta_{lk}$，（正交性）

- $(\langle u_j| \otimes \langle v_l|)|u_i\rangle = \langle u_j|u_i\rangle\langle v_l| = \delta_{ji}\langle v_l|$

· 外积：

$$\left(|u_j\rangle \otimes |v_l\rangle \right)\left(\langle u_i| \otimes \langle v_k| \right)$$

$$= \left(|u_j\rangle\langle u_i| \right) \otimes \left(|v_l\rangle\langle v_k| \right)$$

· 完备性：

$$\sum_i \sum_k \left(|u_i\rangle \otimes |v_k\rangle \right)\left(\langle u_i| \otimes \langle v_k| \right)$$

$$= \left(\sum_i |u_i\rangle\langle u_i| \right) \otimes \left(\sum_k |v_k\rangle\langle v_k| \right)$$

$$= \mathbb{1}(1) \otimes \mathbb{1}(2) \equiv \mathbb{1}(1,2)$$

·

$$\hat{A}(1) \otimes \hat{B}(2) \left[|u_i(1)\rangle \otimes |v_k(2)\rangle \right]$$

$$= \left[\hat{A}(1) |u_i(1)\rangle \right] \otimes \left[\hat{B}(2) |v_k(2)\rangle \right]$$

· 矩阵元：

$$\langle u_j| \otimes \langle v_k| \left[\hat{A} \otimes \hat{B}\right] |u_i\rangle \otimes |v_k\rangle$$

$$= \langle u_j| \hat{A} |u_i\rangle \langle v_k| \hat{B} |v_k\rangle$$

· 部分求迹

$$Tr_1\left[\hat{A}(1) \otimes \hat{B}(2)\right] = Tr_1\left[\hat{A}(1)\right] \hat{B}(2)$$

$$Tr_2\left[\hat{A}(1) \otimes \hat{B}(2)\right] = Tr_2\left[\hat{B}(2)\right] \hat{A}(1)$$

· $\hat{A}(1) \Rightarrow \hat{A}(1) \otimes \mathbb{1}(2)$

· $\hat{B}(2) \Rightarrow \mathbb{1}(1) \otimes \hat{B}(2)$

* $Tr_1\left[\hat{A}(1) \otimes \hat{B}(2)\right] = \sum_i \langle u_i(1)| \hat{A}(1) \otimes \hat{B}(2) |u_i(1)\rangle$

$$= \sum_i \langle u_i(1)| \hat{A}(1) |u_i(1)\rangle \hat{B}(2)$$

- 纯态复合系统态矢量 $|\psi(1,2)\rangle$

- $|\psi(1,2)\rangle = \mathbb{1}\,|\psi(1,2)\rangle$

$$= \sum_i \sum_k \left(|u_i\rangle \otimes |v_k\rangle\right)\left(\langle u_i| \otimes \langle v_k|\right)|\psi(1,2)\rangle$$

$$= \sum_i \sum_k C_{ik}(1,2)\left(|u_i\rangle \otimes |v_k\rangle\right) .$$

($\left(\langle u_i| \otimes \langle v_k|\right)|\psi(1,2)\rangle = C_{ik}(1,2)$)

- $\langle\psi(1,2)|\psi(1,2)\rangle = 1 \Rightarrow$

$$\sum_i \sum_k |C_{ik}(1,2)|^2 = 1$$

- $|\psi(1,2)\rangle$ 可分为两大类：

(1) $C_{ik}(1,2) = C_i(1)\,\tilde{C}_k(2) \Rightarrow$

$$|\psi(1,2)\rangle = \left(\sum_i C_i(1)|u_i(1)\rangle\right) \otimes \left(\sum_k \tilde{C}_k(2)|v_k(2)\rangle\right)$$

$$= |\tilde{\varphi}(1)\rangle \otimes |\tilde{\varphi}(2)\rangle \quad \text{(可分离态)}$$

(2) $C_{ik}(1,2) \neq C_i(1)\widetilde{C}_k(2)$ （纠缠态）

Entangled state

· 纯态复合系统的密度算符 $\hat{\rho}(1,2)$

$$\hat{\rho}(1,2) = |\psi(1,2)\rangle\langle\psi(1,2)| \quad .$$

· 密度矩阵 $\rho(1,2)$

$$\hat{\rho}(1,2) = |\psi(1,2)\rangle\langle\psi(1,2)|$$

$$= \left(\sum_i\sum_k C_{ik}|u_i\rangle\otimes|v_k\rangle\right)\left(\sum_j\sum_l C_{jl}^*\langle u_j|\otimes\langle v_l|\right)$$

$$= \sum_i\sum_k\sum_j\sum_l C_{ik}C_{jl}^*\left(|u_i\rangle\langle u_j|\right)\otimes\left(|v_k\rangle\langle v_l|\right)$$

$$\Rightarrow \quad \rho(1,2) = \begin{bmatrix} \vdots & \\ \cdots & C_{ik}C_{jl}^* \cdots \\ \vdots & \end{bmatrix}$$

$(N\times M)\times(N\times M)$ 矩阵

· 物理量的平均值

$$\overline{\hat{A}(1)\otimes\hat{B}(2)} = \langle \psi(1,2)|\ \hat{A}(1)\otimes\hat{B}(2)\ |\psi(1,2)\rangle$$

$$= \sum_{ik} \langle \psi(1,2)|\ \hat{A}(1)\otimes\hat{B}(2)\left(|u_i\rangle\otimes|v_k\rangle\right)\left(\langle u_i|\otimes\langle v_k|\right)|\psi(1,2)\rangle$$

$$= \sum_{ik} \langle u_i|\otimes\langle v_k|\ \psi(1,2)\rangle\langle\psi(1,2)|\ \hat{A}(1)\otimes\hat{B}(2)\ |u_i\rangle\otimes|v_k\rangle$$

$$= \sum_{ik} \langle u_i|\otimes\langle v_k|\ \hat{\rho}(1,2)\ \hat{A}(1)\otimes\hat{B}(2)\ |u_i\rangle\otimes|v_k\rangle$$

$$= \mathrm{Tr}\left[\hat{\rho}(1,2)\ \hat{A}(1)\otimes\hat{B}(2)\right]$$

即

$$\overline{\hat{A}(1)\otimes\hat{B}(2)} = \mathrm{Tr}\left[\hat{\rho}(1,2)\ \hat{A}(1)\otimes\hat{B}(2)\right]$$

· 例：

复合系统：2个电子自旋系统

$$\begin{cases} \text{单一个子系统基矢：} & |\uparrow\rangle_1,\ |\downarrow\rangle_1 \Rightarrow \{|u_i(1)\rangle\} \\ \text{单二个子系统基矢：} & |\uparrow\rangle_2,\ |\downarrow\rangle_2 \Rightarrow \{|v_k(2)\rangle\} \end{cases}$$

各二维

复合系统基矢之一.（无耦合表象）

（略去直积记号 $\otimes$）

$$\begin{cases} |\chi_1\rangle \equiv |\uparrow\rangle_1 |\uparrow\rangle_2 \\ |\chi_2\rangle \equiv |\uparrow\rangle_1 |\downarrow\rangle_2 \\ |\chi_3\rangle \equiv |\downarrow\rangle_1 |\uparrow\rangle_2 \\ |\chi_4\rangle \equiv |\downarrow\rangle_1 |\downarrow\rangle_2 \end{cases} \Rightarrow \{|u_i(1)\rangle \otimes |v_k(2)\rangle\}$$

四维.

$$\begin{cases} \langle\chi_i|\chi_j\rangle = \delta_{ij} \\ \sum_{i=1}^{4} |\chi_i\rangle\langle\chi_i| = \mathbb{1} \end{cases}$$

（各为可分离态）

· 复合系统基矢的矩阵表式

$|A\rangle_1 \Rightarrow \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$, $|B\rangle_2 \Rightarrow \begin{pmatrix} B_1 \\ B_2 \end{pmatrix}$, 则

· $|A\rangle_1 \otimes |B\rangle_2 \Rightarrow \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} \otimes \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = \begin{pmatrix} A_1 B_1 \\ A_1 B_2 \\ A_2 B_1 \\ A_2 B_2 \end{pmatrix}$

· 第一子系统基矢：

$|\uparrow\rangle_1 \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}_1$, $|\downarrow\rangle_1 \Rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}_1$

· 第二子系统基矢：

$|\uparrow\rangle_2 \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}_2$, $|\downarrow\rangle_2 \Rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}_2$ 则

· $|\chi_1\rangle = |\uparrow\rangle_1 |\uparrow\rangle_2 \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}_1 \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

· $|\chi_2\rangle = |\uparrow\rangle_1 |\downarrow\rangle_2 \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}_1 \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

- $|\chi_3\rangle = |\downarrow\rangle_1 |\uparrow\rangle_2 \Rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}_1 \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$,

- $|\chi_4\rangle = |\downarrow\rangle_1 |\downarrow\rangle_2 \Rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}_1 \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

- 复合系统算符的矩阵表式

- 为一子系统算符：

$$\hat{F}(1) \Rightarrow \begin{pmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{pmatrix}$$

- 为二子系统算符

$$\hat{G}(2) \Rightarrow \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix}$$ 则

$$\hat{F}(1) \otimes \hat{G}(2) \Rightarrow \begin{pmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{pmatrix} \otimes \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix} =$$

$$\begin{pmatrix} F_{11}(G) & F_{12}(G) \\ F_{21}(G) & F_{22}(G) \end{pmatrix} \otimes \begin{pmatrix} G_{11}(G) & G_{12}(G) \\ G_{21}(G) & G_{22}(G) \end{pmatrix}$$

其中，

$$= \begin{pmatrix} F_{11}G_{11} & F_{11}G_{12} & F_{12}G_{11} & F_{12}G_{12} \\ F_{11}G_{21} & F_{11}G_{22} & F_{12}G_{21} & F_{12}G_{22} \\ F_{21}G_{11} & F_{21}G_{12} & F_{22}G_{11} & F_{22}G_{12} \\ F_{21}G_{21} & F_{21}G_{22} & F_{22}G_{21} & F_{22}G_{22} \end{pmatrix}.$$

例：$\hat{S}_x(1) \otimes \hat{S}_y(2)$ 的矩阵表示

$$\hat{S}_x(1) \Rightarrow \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_y(2) \Rightarrow \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{S}_x(1) \otimes \hat{S}_y(2) \Rightarrow \frac{\hbar^2}{4}\begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}.$$

※ 复合系统基矢之二 （耦合表象）

- $|\hat{\chi}_1\rangle \equiv |\uparrow\rangle_1 |\uparrow\rangle_2$
- $|\hat{\chi}_2\rangle \equiv |\downarrow\rangle_1 |\downarrow\rangle_2$

} 自旋三重态

- $|\hat{\chi}_3\rangle \equiv \frac{1}{\sqrt{2}} \left[|\uparrow\rangle_1 |\downarrow\rangle_2 + |\downarrow\rangle_1 |\uparrow\rangle_2 \right]$
- $|\hat{\chi}_4\rangle \equiv \frac{1}{\sqrt{2}} \left[|\uparrow\rangle_1 |\downarrow\rangle_2 - |\downarrow\rangle_1 |\uparrow\rangle_2 \right]$ 自旋单态

（正交.完备）

- $|\hat{\chi}_1\rangle$, $|\hat{\chi}_2\rangle$ 为可分离态.

  $|\hat{\chi}_3\rangle$, $|\hat{\chi}_4\rangle$ 为纠缠态.

4 复合系统基元之三（Bell基）

Bell基

$$|\psi_{Bi}\rangle \propto (|0\rangle_1, |1\rangle_1)\,\sigma_i \begin{pmatrix} |0\rangle_2 \\ |1\rangle_2 \end{pmatrix}$$

$$i = 0, 1, 2, 3$$

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$|\psi_{B0}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_1|0\rangle_2 + |1\rangle_1|1\rangle_2),$$

$$|\psi_{B1}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_1|1\rangle_2 + |1\rangle_1|0\rangle_2),$$ 正交完备

$$|\psi_{B2}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_1|1\rangle_2 - |1\rangle_1|0\rangle_2),$$ → EPR对

$$|\psi_{B3}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_1|0\rangle_2 - |1\rangle_1|1\rangle_2).$$ 量子力学的非局域性

（$\sigma_1^*\sigma_2$, $\sigma_1^3\sigma_2^3$, $\sigma_1^2\sigma_2^2$ 的共同本征态，皆为纠缠态）

例：$|\psi(1,2)\rangle = \frac{\sqrt{2}}{2}|\chi_1\rangle + \frac{1}{2}|\chi_2\rangle + \frac{1}{2}|\chi_4\rangle$

（纠缠态）

· 密度算符

$$\hat{\rho}(1,2) = |\psi(1,2)\rangle\langle\psi(1,2)|$$

$$= \frac{1}{2}|\chi_1\rangle\langle\chi_1| + \frac{\sqrt{2}}{4}|\chi_1\rangle\langle\chi_2| + \frac{\sqrt{2}}{4}|\chi_1\rangle\langle\chi_4|$$

$$+ \frac{\sqrt{2}}{4}|\chi_2\rangle\langle\chi_1| + \frac{1}{4}|\chi_2\rangle\langle\chi_2| + \frac{1}{4}|\chi_2\rangle\langle\chi_4|$$

$$+ \frac{\sqrt{2}}{4}|\chi_4\rangle\langle\chi_1| + \frac{1}{4}|\chi_4\rangle\langle\chi_2| + \frac{1}{4}|\chi_4\rangle\langle\chi_4|.$$

· 密度矩阵　$\rho_{ij}(1,2) \equiv \langle\chi_i|\hat{\rho}(1,2)|\chi_j\rangle$

$$\rho(1,2) = \begin{array}{c} \\ \chi_1 \\ \chi_2 \\ \chi_3 \\ \chi_4 \end{array} \begin{array}{c} \begin{array}{cccc} \chi_1 & \chi_2 & \chi_3 & \chi_4 \end{array} \\ \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}}{4} \\ \frac{\sqrt{2}}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & \frac{1}{4} & 0 & \frac{1}{4} \end{pmatrix} \end{array}$$

另一种求法：

第一项：$\frac{1}{2}|x_1\rangle\langle x_1| \Rightarrow$

$$\frac{1}{2}\begin{pmatrix}1\\0\\0\\0\end{pmatrix}(1\ 0\ 0\ 0) = \frac{1}{2}\begin{pmatrix}1&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{pmatrix},$$

第二项：$\frac{1}{4}|x_1\rangle\langle x_2| \Rightarrow$

$$\frac{1}{4}\begin{pmatrix}1\\0\\1\\0\end{pmatrix}(0\ 1\ 0\ 0) = \frac{1}{4}\begin{pmatrix}0&1&0&0\\0&0&0&0\\0&1&0&0\\0&0&0&0\end{pmatrix},$$

$$\vdots$$

· 九个项相加，可得密度矩阵.

$$\rho(1,2) = \begin{pmatrix}\frac{1}{2}&\frac{1}{4}&0&\frac{1}{4}\\\frac{1}{4}&\frac{1}{4}&0&\frac{1}{4}\\0&0&0&0\\\frac{1}{4}&\frac{1}{4}&0&\frac{1}{4}\end{pmatrix}$$

§2.8 约化密度算符和密度矩阵

- 求 $\hat{B}(2)$ 的平均值

$$\overline{\hat{B}(2)} = \langle \psi(1,2) | \mathbb{1}(1) \otimes \hat{B}(2) | \psi(1,2) \rangle$$

$$= \mathrm{Tr}\left[\hat{\rho}(1,2)\, \mathbb{1}(1) \otimes \hat{B}(2) \right] \quad \cdots (1)$$

- $\overline{\hat{B}(2)} \neq \sum_k \langle \psi_k | \underbrace{\hat{\rho}_R(2)}\, \hat{B}(2) | \psi_k \rangle = \mathrm{Tr}_2 [\underbrace{\hat{\rho}_R(2)}\, \hat{B}(2)]$

$\cdots (2)$

可以！

以下，从 (1) 式出发，导出 (2) 式中的

$\hat{\rho}_R(2)$ 的具体形式。

- $\langle \Psi(1,2) | \mathbb{1}(1) \otimes \hat{B}(2) | \Psi(1,2) \rangle$

$$= \left\{ \sum_{ik} C_{ik}^{*} \langle u_i | \otimes \langle v_k | \right\} \mathbb{1}(1) \otimes \hat{B}(2) \left\{ \sum_{jl} C_{jl} | u_j \rangle \otimes | v_l \rangle \right\}$$

$$= \sum_{ik} \sum_{jl} C_{ik}^{*} C_{jl} \langle u_i | u_j \rangle \langle v_k | \hat{B}(2) | v_l \rangle$$

($\langle u_i | u_j \rangle = \delta_{ij}$)

$$= \sum_{ik} \sum_{l} C_{ik}^{*} C_{il} \langle v_k | \hat{B}(2) | v_l \rangle$$

$$= \sum_{ik} \sum_{lj} C_{ik}^{*} C_{il} \langle v_k | \hat{B}(2) | v_j \rangle \langle v_j | v_l \rangle$$

$$= \sum_{ik} \sum_{lj} C_{ik}^{*} C_{il} \langle v_j | v_l \rangle \langle v_k | \hat{B}(2) | v_j \rangle$$

$$= \sum_{j} \langle v_j | \left[\sum_{ik} \sum_{l} C_{ik}^{*} C_{il} | v_l \rangle \langle v_k | \right] \hat{B}(2) | v_j \rangle$$

$$\equiv \mathrm{Tr}_2 \left[\hat{\rho}_R(2) \hat{B}(2) \right],$$

$$\Rightarrow \quad \hat{\rho}_R(2) = \sum_{ik} \sum_{l} C_{ik}^{*} C_{il} | v_l \rangle \langle v_k |$$

{ 求二子系统 约化密度 算符

类似，

$$\overline{\hat{A}^{(1)}} = Tr_1\left[\hat{\rho}_R^{(1)}\hat{A}^{(1)}\right], \quad \text{其中}$$

$$\hat{\rho}_R^{(1)} = \sum_{ik}\sum_{j} c_{ik}^{*} c_{jk} |u_j\rangle\langle u_i| .$$

{ 第一子系统约化密度算符.

易证：

$$\begin{cases} Tr_1\left[\hat{\rho}_R^{(1)}\right] = 1 \ ; & Tr_1\left[\hat{\rho}_R^{2(1)}\right] \leq 1 \\ Tr_2\left[\hat{\rho}_R^{(2)}\right] = 1 \ ; & Tr_2\left[\hat{\rho}_R^{2(2)}\right] \leq 1 \end{cases}$$

等式对应于 $|\psi(12)\rangle$ 为可分离态的情况，（纯态密度算符）

不等式对应于纠缠态情况。（混合态密度算符）

- 计算 $Tr_1[\hat{\rho}(1,2)] \neq \hat{\rho}_R(2)$

- $Tr_1[\hat{\rho}(1,2)] = \sum_n \langle u_n | \hat{\rho}(1,2) | u_n \rangle$

$$= \sum_n \langle u_n | \psi(1,2) \rangle \langle \psi(1,2) | u_n \rangle$$

$$= \sum_n \langle u_n | \left[\sum_{\substack{ik \\ jl}} c_{ik}^* c_{jl} \, |u_j\rangle \otimes |v_l\rangle \langle u_i| \otimes \langle v_k| \right] | u_n \rangle$$

$$= \sum_{ik} \sum_{jl} c_{ik}^* c_{jl} \sum_n \langle u_n | u_j \rangle \langle u_i | u_n \rangle |v_l\rangle\langle v_k|$$

($\langle u_n|u_j\rangle = \delta_{nj}$, $\langle u_i|u_n\rangle = \delta_{in}$)

$$= \sum_{ikl} c_{ik}^* c_{il} \, |v_l\rangle\langle v_k| = \hat{\rho}_R(2)$$ (压缩算符)

即，$\hat{\rho}_R(2) = Tr_1[\hat{\rho}(1,2)]$

类似，$\hat{\rho}_R(1) = Tr_2[\hat{\rho}(1,2)]$

上例：
- 单一子系统基矢：$|\uparrow\rangle_1$，$|\downarrow\rangle_1$
- 单一子系统基矢：$|\uparrow\rangle_2$，$|\downarrow\rangle_2$
- 复合系统基矢：$|\chi_1\rangle, |\chi_2\rangle, |\chi_3\rangle, |\chi_4\rangle$

$$\hat{\rho}(1,2) = \frac{1}{2}|\chi_1\rangle\langle\chi_1| + \frac{\sqrt{2}}{4}|\chi_1\rangle\langle\chi_2| + \frac{\sqrt{2}}{4}|\chi_1\rangle\langle\chi_4|$$

$$+ \frac{\sqrt{2}}{4}|\chi_2\rangle\langle\chi_1| + \frac{1}{4}|\chi_2\rangle\langle\chi_2| + \frac{1}{4}|\chi_2\rangle\langle\chi_4|$$

$$+ \frac{\sqrt{2}}{4}|\chi_4\rangle\langle\chi_1| + \frac{1}{4}|\chi_4\rangle\langle\chi_2| + \frac{1}{4}|\chi_4\rangle\langle\chi_4|$$

$$\hat{\rho}_R(1) = Tr_2[\hat{\rho}(1,2)] = {}_2\langle\uparrow|\hat{\rho}(1,2)|\uparrow\rangle_2 + {}_2\langle\downarrow|\hat{\rho}(1,2)|\downarrow\rangle_2$$

$$= \frac{3}{4}|\uparrow\rangle_1\langle\uparrow| + \frac{1}{4}|\uparrow\rangle_1\langle\downarrow| + \frac{1}{4}|\downarrow\rangle_1\langle\uparrow| + \frac{1}{4}|\downarrow\rangle_1\langle\downarrow|$$

$$\Rightarrow \rho_R(1) = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

$$\left(Tr_1[\hat{\rho}_R(1)] = 1 \right)$$

$$\left(Tr_1[\hat{\rho}_R^2(1)] = \frac{3}{4} < 1 \right)$$

$\Rightarrow |\psi(1,2)\rangle$ 是纠缠态！

• 求约化密度矩阵

例 2 : $|\psi(1,2)\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_1 \otimes |\downarrow\rangle_2 - |\downarrow\rangle_1 \otimes |\uparrow\rangle_2 \right)$

EPR - 对

• 求 $\hat{\rho}_R(1)$ 及 $\rho_R(1)$

$$\hat{\rho}(1,2) = |\psi(1,2)\rangle\langle\psi(1,2)|$$

$$= \left\{ \frac{1}{\sqrt{2}} \left(|\uparrow\rangle_1 \otimes |\downarrow\rangle_2 - |\downarrow\rangle_1 \otimes |\uparrow\rangle_2 \right) \right\}$$

$$\left\{ \frac{1}{\sqrt{2}} \left({}_2\langle\downarrow| \otimes {}_1\langle\uparrow| - {}_2\langle\uparrow| \otimes {}_1\langle\downarrow| \right) \right\}$$

$$= \frac{1}{2} \Big\{ |\uparrow\rangle_1\langle\uparrow| \otimes |\downarrow\rangle_{22}\langle\downarrow| - |\uparrow\rangle_{11}\langle\downarrow| \otimes |\downarrow\rangle_2\langle\uparrow|$$

$$- |\downarrow\rangle_{11}\langle\uparrow| \otimes |\uparrow\rangle_{22}\langle\downarrow| + |\downarrow\rangle_{11}\langle\downarrow| \otimes |\uparrow\rangle_{22}\langle\uparrow| \Big\}$$

$$\Rightarrow \quad \hat{\rho}_R(1) = \mathrm{Tr}_2\left[\hat{\rho}(1,2)\right]$$

$$= \langle\uparrow|_2\, \hat{\rho}(1,2)\, |\uparrow\rangle_2 + \langle\downarrow|_2\, \hat{\rho}(1,2)\, |\downarrow\rangle_2$$

$$= \frac{1}{2}\,|\downarrow\rangle_{11}\langle\downarrow| + \frac{1}{2}\,|\uparrow\rangle_{11}\langle\uparrow|$$

$$\Rightarrow \quad \rho_R(1) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

混合态密度矩阵 → $\hat{\rho}(1,2)$ 为纠缠态密度算符！

即 $|\psi(1,2)\rangle$ 为纠缠态.

类似可得：

$$\hat{\rho}_R(2) = \frac{1}{2}\,|\uparrow\rangle_2\langle\uparrow| + \frac{1}{2}\,|\downarrow\rangle_2\langle\downarrow|$$

$$\Rightarrow \quad \rho_R(2) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

§2.9 复合系统纯态的 Schmidt 分解

decomposition

- 复合系统包含两个子系统

$$\begin{cases} \text{第一子系统：} & \{|u_i(1)\rangle\} \quad N\text{-维} \\ \text{第二子系统：} & \{|v_k(2)\rangle\} \quad M\text{-维} \end{cases}$$

$\Rightarrow$ 复合系统：$N \times M$ 维

基矢：$\{|u_i(1)\rangle \otimes |v_k(2)\rangle\}$

- 纯态 $|\psi(1,2)\rangle = \sum_{ik} C_{ik} |u_i(1)\rangle \otimes |v_k(2)\rangle$

一般含 $N \times M$ 个项！

- Schmidt 分解

$$|\psi(1,2)\rangle = \sum_{i=1}^{N_s} \sqrt{p_i}\, |\tilde{u}_i(1)\rangle \otimes |\tilde{v}_i(2)\rangle$$

其中，$\begin{cases} N_s \leq N, M, \\ \sum_{i=1}^{N_s} p_i = 1 \end{cases}$ $(0 < p_i \leq 1)$

而，$\{|\tilde{u}_i(1)\rangle\}$ 和 $\{|\tilde{v}_i(2)\rangle\}$ 分别为各子系统的另一套正交归一矢量集.

证明：（假设 $N<M$）

① 由给定的 $|\psi(1,2)\rangle$ 可得复合系统密度算符的表达式：$\hat{\rho}(1,2)=|\psi(1,2)\rangle\langle\psi(1,2)|$.

② 由 $\hat{\rho}(12)$ 得，$\hat{\rho}_R(1)=Tr_2[\hat{\rho}(1,2)]$，以及矩阵 $\rho_R(1)$. （$N\times N$ 方矩阵）

③ 对子系统进行表象变换，变换到 $\hat{\rho}_R(1)$ 的本征表象，即

$$\{u_i(1)\} \longrightarrow \{\tilde{u}_i(1)\},$$

其中，$|\tilde{u}_i(1)\rangle$ 为 $\hat{\rho}_R(1)$ 的本征值为 ρ_i 的本征态：

$$\hat{\rho}_R(1)|\tilde{u}_i(1)\rangle=\rho_i|\tilde{u}_i(1)\rangle$$

· 可以求出一组正交矩阵

$$U = \begin{bmatrix} U_{11} & U_{12} & \cdots & U_{1N} \\ & \vdots & & \\ U_{N1} & U_{N2} & \cdots & U_{NN} \end{bmatrix}$$

将矩阵 $R_R(1)$ 对角化：（$N \times N$ 矩阵）

$$\begin{pmatrix} U_{11} & U_{12} & \cdots & U_{1N} \\ & \vdots & & \\ U_{N1} & U_{N2} & \cdots & U_{NN} \end{pmatrix}^{+} \begin{pmatrix} R_R^{(1)}(1,1) & R_R^{(1)}(1,2) & \cdots & R_R^{(1)}(1,N) \\ & \vdots & & \\ R_R^{(1)}(N,1) & R_R^{(1)}(N,2) & \cdots & R_R^{(1)}(N,N) \end{pmatrix}$$

$\searrow R_R(1)$

$$\times \begin{pmatrix} U_{11} & U_{12} & \cdots & U_{N1} \\ & \vdots & & \\ U_{N1} & U_{N2} & \cdots & U_{NN} \end{pmatrix} = \begin{pmatrix} P_1 & & & 0 \\ & P_2 & & \\ & & \ddots & \\ 0 & & & P_N \end{pmatrix}$$

设其中不为零的本征值（对角元）共有 N_s 个，则

必有 $\sum_{i=1}^{N_s} P_i = 1$ 。（密度矩阵的性质）

（$N_s \leq N$）

※ 变换矩阵 U 以及本征值 P_i 都可以通过求解 $R_R(1)$ 的本征值与矢量得到。

· 基矢间的变换关系为：

正交·完备

$$\begin{pmatrix} |u_1(1)\rangle \\ \vdots \\ |u_N(1)\rangle \end{pmatrix} = \begin{pmatrix} U_{11}^* & \cdots & U_{1N}^* \\ & \vdots & \\ U_{N1}^* & \cdots & U_{NN}^* \end{pmatrix} \begin{pmatrix} |\tilde{u}_1(1)\rangle \\ \vdots \\ |\tilde{u}_N(1)\rangle \end{pmatrix},$$

即

$$|u_i(1)\rangle = \sum_{j=1}^{N} U_{ij}^* |\tilde{u}_j(1)\rangle \quad (i=1,2,\cdots N).$$

· 在新表象中，有

$$\hat{\rho}_R(1) = \sum_{i=1}^{N_s} p_i \, |\tilde{u}_i(1)\rangle\langle\tilde{u}_i(1)|$$

（通表示）

④ 将 $|u_i(1)\rangle = \sum_{j=1}^{N} U_{ij}^{*} |\tilde{u}_j(1)\rangle$ 代入

$$|\psi(1,2)\rangle = \sum_{ik} c_{ik} |u_i(1)\rangle \otimes |v_k(2)\rangle$$

$$\Rightarrow |\psi(1,2)\rangle = \sum_{j=1}^{N} |\tilde{u}_j(1)\rangle \otimes \left(\sum_{ik} U_{ij}^{*} c_{ik} \right) |v_k(2)\rangle$$

令 $\sum_{i=1}^{N} U_{ij}^{*} c_{ik} \equiv d_{jk}$

$$\Rightarrow |\psi(1,2)\rangle = \sum_{j=1}^{N} |\tilde{u}_j(1)\rangle \otimes \sum_{k=1}^{M} d_{jk} |v_k(2)\rangle$$

再令 $\sum_{k=1}^{M} d_{jk} |v_k(2)\rangle \equiv \sqrt{p_j}\, |\tilde{v}_j(2)\rangle$

$$\Rightarrow |\psi(1,2)\rangle = \sum_{j=1}^{N_S} \sqrt{p_j}\, |\tilde{u}_j(1)\rangle \otimes |\tilde{v}_j(2)\rangle$$

⑤ $\langle \tilde{v}_j(2) | \tilde{v}_i(2) \rangle \neq \delta_{ij}$

$$|\tilde{v}_j(2)\rangle \equiv \sum_{k=1}^{M} d_{jk} |v_k(2)\rangle / \sqrt{p_j}$$

一方で:

$$|\psi(12)\rangle = \sum_{j=1}^{N_s} |\widetilde{u_j}(1)\rangle \otimes \sum_{k=1}^{M} d_{jk} |v_k(2)\rangle$$

$$\hat{\rho}(1,2) = |\psi(1,2)\rangle\langle\psi(1,2)|$$

$$= \left(\sum_j |\widetilde{u_j}(1)\rangle \otimes \sum_k d_{jk} |v_k(2)\rangle \right)$$

$$\left(\sum_{j'} \langle \widetilde{u_{j'}}(1)| \otimes \sum_{k'} d^*_{j'k'} \langle v_{k'}(2)| \right)$$

- $\hat{\rho}_R(1) = \mathrm{Tr}_2[\hat{\rho}(1,2)] = \sum_{R=1}^{M} \langle v_R(2)| \hat{\rho}(1,2) |v_R(2)\rangle$

$$= \sum_I \sum_{\substack{k \\ k'}} \sum_j \sum_{j'} |\widetilde{u_j}(1)\rangle\langle\widetilde{u_{j'}}(1)| d_{jk} d^*_{j'k'} \delta_{Rk} \delta_{Rk'}$$

$$= \sum_{R=1}^{M} \sum_{j=1}^{N_s} \sum_{j'=1}^{N_s} |\widetilde{u_j}(1)\rangle\langle\widetilde{u_{j'}}(1)| d_{jR} d^*_{j'R} .$$